

Sai Sirisha Madishetty

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PROFESSIONAL SUMMARY

- Developed and optimized end-to-end **software applications** and **data engineering** solutions with **Python, MySQL, and Flask**, driving efficient data processing and scalable architecture.
- Architected and deployed advanced data solutions using **Big Data technologies** such as **Apache Hadoop** and **Data Warehousing**, transforming raw data into actionable insights and scalable models.
- Demonstrated a proven track record of delivering high-quality software solutions within both **Agile** and **Waterfall** environments, leveraging project management tools like **JIRA** and ServiceNow to ensure seamless communication and timely delivery.
- Leveraged AWS services, including **IAM, EC2, S3, Glue, Redshift, and Athena**, to build, manage, and optimize robust data pipelines and analytics workflows, enhancing data accessibility and security.
- Applied **Machine Learning** algorithms and Python libraries, such as **scikit-learn** and **TensorFlow**, to conduct data analysis and predictive modeling, contributing to data-driven decision-making processes.
- Collaborated cross-functionally to design and implement **RESTful APIs** and **microservices** with Flask, ensuring seamless data integration and system scalability across various platforms.
- Experienced in managing and optimizing **Azure data services**, including **Azure Data Factory, Azure SQL Database, and Azure Blob Storage**, to support data integration and scalable storage solutions.
- Pioneered data quality frameworks and monitoring tools using **Python** and **AWS CloudWatch**, elevating data reliability and operational efficiency in a cloud environment.

TECHNICAL SKILLS

Programming Languages: Python, Go, R, Java, C, C++, JavaScript.

Cloud Technologies: Aws (EC2, S3, Lambda, Athena, Redshift), Microsoft Azure(Azure Data Factory, Azure Blob Storage).

Data Analysis Libraries: Pandas, NumPy, TensorFlow, Scikit-Learn, Seaborn, OpenCV, PyTorch.

Machine Learning Models: Linear Regression, Logistic Regression, Principal Component Analysis (PCA), Natural Language Processing (NLP), K-Means Clustering.

Deep Learning Models: Convolutional Neural Networks (CNN).

Frameworks: SpringBoot, Power Mockito

Data Engineering: Data Warehousing, ETL Tools, Apache Spark.

Database Management Systems: MySQL, NoSQL

Data Visualization: Tableau, PowerBI, SAP Analytics

WORK EXPERIENCE

Lensoft Solutions

November 2024 – Till Date

Role : Junior Data Analyst

Responsibilities:

- Analyzing large datasets using **SQL** to extract actionable insights and support decision-making processes.
- Developing interactive dashboards and reports using **Power BI, Tableau, and SAP Analytics Cloud** to visualize data effectively.
- Managing and maintaining cloud-based data pipelines and storage solutions using **AWS** services such as **S3, RDS, and Redshift**.
- Collaborating with cross-functional teams to gather business requirements and translate them into data solutions.
- Ensuring data integrity and accuracy by performing data cleansing, validation, and quality checks.
- Automating routine data tasks and workflows using **Python** or other scripting tools to enhance efficiency.
- Utilizing **SAP** modules to integrate, analyze, and report on business processes for improved efficiency.
- Documenting methodologies, processes, and findings to create accessible resources for team members and stakeholders.
- Integrating data from SAP systems with other platforms to create unified dashboards and reports for comprehensive analysis.
- Monitoring and troubleshooting data pipelines and cloud-based workflows to ensure seamless operations.
- Supporting data migration and transformation projects involving SAP systems and **AWS** services for enhanced data accessibility.

Environment: SQL, AWS (S3, RDS, Redshift), Power BI, Tableau, SAP Analytics Cloud, SAP modules, Python, data visualization, data cleansing, data validation, cloud technologies, business intelligence tools.

KP IT Solutions LLC- Austin,Texas

May 2024 – November 2024

Role: Programmer Analyst

Responsibilities:

- Engineer and deploy high-performance microservices and **RESTful APIs** using **Spring Boot** on cloud platforms like **AWS**, ensuring scalable and robust solutions tailored to business needs.
- Architect and implement a microservices-based architecture, promoting seamless integration and interoperability across various services and enhancing system modularity and maintainability.
- Author and optimize **JUnit test** cases with **Power Mockito**, executing peer-level testing and **behavior-driven development (BDD)** to elevate code coverage, ensure compliance with quality standards, and achieve seamless software validation.
- Adopt and champion the **Agile (Scrum)** methodology for application development, actively participating in sprint planning, backlog grooming, and retrospectives to foster continuous improvement and team collaboration.
- Manage and streamline the deployment pipeline and release management process, leveraging cloud infrastructure and implementing automation with **CI/CD** practices using Jenkins to enable continuous delivery and enhance release efficiency.
- Collaborate closely with cross-functional teams to gather requirements, prioritize features, and deliver solutions that align with business objectives, ensuring that all components of the architecture are designed for maximum scalability and resilience.
- Conduct performance optimization on existing applications, performing code refactoring, implementing caching mechanisms, and utilizing **AWS** monitoring tools like **CloudWatch** to reduce latency and improve response times, ultimately enhancing the end-user experience.

Environments: Java, Spring Boot, JUnit, Power Mockito, AWS (including EC2, S3, CloudWatch, Lambda, RDS, Elastic Beanstalk), Jenkins, Agile (Scrum), Behavior-Driven Development (BDD), Microservices Architecture, RESTful APIs, CI/CD, Cloud Infrastructure, Peer-level Testing, Automated Deployment and Release Management

Cognizant Technology Solutions - Hyderabad, India

September 2021 – August 2022

Role: Python Developer

Responsibilities:

- Leveraged **AWS services** such as **S3** for scalable data storage, **Glue** for ETL processes, **Athena** for querying large datasets, **ECS** for containerized application management, and **SageMaker** for training machine learning models, enhancing data-driven solutions and analytical capabilities.
- Developed and optimized ETL processes using **PySpark** and **Spark SQL** for efficient data extraction, transformation, and loading, collaborating with data scientists and analysts to streamline data wrangling and transformation workflows in **AWS Glue**.
- Automated **ETL workflows** in **AWS Glue**, utilizing advanced ETL pipelines and cloud-native services to ensure high data integrity and availability across multiple data sources.
- Integrated **diverse data sources** into scalable, production-ready solutions, improving data maturity and aligning with key business analytics initiatives for actionable insights.
- Implemented **Agile methodologies** to support continuous integration and delivery of data engineering solutions, facilitating seamless collaboration with cross-functional teams and maintaining a high standard of code quality and performance.
- Developed **data validation and quality checks** using **Python** and AWS-native tools to ensure accuracy, consistency, and reliability of data across all stages of the ETL pipeline, helping to detect and resolve data issues early.
- Optimized query performance and data retrieval times by tuning **Spark** configurations and caching frequently used datasets in **AWS Glue**, resulting in faster processing and improved data accessibility for end-users.

Environments: Python, PySpark, AWS (including S3, Glue, Athena, ECS, SageMaker), Agile, Continuous Integration and Delivery (CI/CD), ETL, Data Wrangling, Data Transformation, Production-ready Pipelines, Data Analytics, Cross-functional Collaboration

PixoPlay IT Services - India

May 2020 – June 2021

Role : Data Analyst

Responsibilities:

- Collected, cleaned, and meticulously analyzed extensive datasets using **Python** and **Matplotlib**, ensuring data accuracy and integrity across all reporting metrics.
- Crafted and optimized complex **SQL** queries to extract, manipulate, and manage data from relational databases, delivering actionable insights to support data-driven decision-making.
- Designed and developed dynamic, interactive dashboards and comprehensive reports in **Tableau**, transforming complex data into visual narratives that highlight key business insights.
- Automated end-to-end data analysis workflows and reporting processes with **Python** scripts, significantly boosting operational efficiency and reducing manual tasks.
- Collaborated cross-functionally with data scientists, analysts, and business stakeholders to understand and solve complex data challenges, tailoring solutions to meet diverse analytical needs.
- Implemented robust data validation and quality checks within the **ETL** processes, leveraging **Pandas** and **SQL**, to uphold data consistency and reliability in all analyses.

Environments: Python, Matplotlib, SQL, Tableau, Pandas, relational databases, ETL processes, data validation, cross-functional collaboration, scripting automation.

CERTIFICATIONS

- **Databricks Generative AI Fundamentals** - [LINK](#)
- **Machine Learning in Python** – Issued by IBM- [LINK](#)

EDUCATION

ROWAN UNIVERSITY

Glassboro, NJ

Master of Science in Data Science

COURSES: Advance Deep Learning Models, Data Mining, Applied Multivariate Data Analysis, Data Warehousing.

VIGNAN INSTITUTE OF TECHNOLOGY AND SCIENCE

Telangana, India

Bachelor of Technology in Computer Science

Courses: Python ,Object-Oriented Programming, Data Structures, Cloud Computing, Operating systems, Machine Learning Algorithms